Linux Users, Groups & Permissions – Basic Task List

Level 1: Users

1. Display the current username.

```
root@LAPTOP-KRQIMQLA:~# whoami
root
```

2. Create a user named pooja

```
root@LAPTOP-KRQIMQLA:~# adduser pooja
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for pooja
Enter the new value, or press ENTER for the default
        Full Name []: Poojaharini
        Room Number []: 1
        Work Phone []: 6383503203
        Home Phone []: 6383503203
        Other []: developer
Is the information correct? [Y/n] y
```

3. Create a user named lilly.

```
Is the information correct? [Y/n] y
root@LAPTOP-KRQIMQLA:~# adduser lilly
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for lilly
Enter the new value, or press ENTER for the default
        Full Name []: Lilly
        Room Number []: 2
        Work Phone []: 9842191591
        Home Phone []: 9842191591
```

4. Create a user named afshaan.

```
root@LAPTOP-KRQIMQLA:~# adduser afshaan
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for afshaan
Enter the new value, or press ENTER for the default
        Full Name []: Afshaan
        Room Number []: 3
        Work Phone []: 9845612398
        Home Phone []: 8954612365
        Other []: developer
Is the information correct? [Y/n] y
```

5. Create a user named soumya.

```
root@LAPTOP-KRQIMQLA:~# adduser soumya
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for soumya
Enter the new value, or press ENTER for the default
        Full Name []: Soumya
        Room Number []: 4
        Work Phone []: 7458963201
        Home Phone []: 7456321089
        Other []: tester
Is the information correct? [Y/n] y
```

6. Display the user ID and group ID of pooja.

```
root@LAPTOP-KRQIMQLA:~# id pooja
uid=1000(pooja) gid=1000(pooja) groups=1000(pooja),100(users)
```

7. Check the groups of the current user.

```
root@LAPTOP-KRQIMQLA:~# groups
root
```

8. Change the login shell of lilly to Bash using usermod.

```
root@LAPTOP-KRQIMQLA:~# usermod -s /bin/bash lilly
usermod: no changes
root@LAPTOP-KRQIMQLA:~# grep '^lilly:' /etc/passwd
lilly:x:1001:1001:Lilly,2,9842191591,9842191591,developer:/home/lilly:/bin/b
ash
```

Level 2: Groups

9. Create a group named `developers`.

```
root@LAPTOP-KRQIMQLA:~# groupadd developers
```

10. Create a group named `testers`.

```
root@LAPTOP-KRQIMQLA:~# groupadd testers
```

```
root@LAPTOP-KRQIMQLA:~# getent group developers testers
developers:x:1004:
testers:x:1005:
```

11. Add `pooja` to the `developers` group.
12. Add `lilly` to the `testers` group.
13. Add `afshaan` to the `developers` group.
14. Add `soumya` to the `testers` group.

```
root@LAPTOP-KRQIMQLA:~# usermod -aG developers pooja
usermod -aG testers lilly
usermod -aG developers afshaan
usermod -aG testers soumya
```

15. Add `pooja` to the `testers` group without removing existing supplementary groups.

```
root@LAPTOP-KRQIMQLA:~# usermod -aG testers pooja
```

16. Display the groups of `pooja`.

```
root@LAPTOP-KRQIMQLA:~# groups pooja
pooja : pooja users developers testers
```

17. Change the primary group of `lilly` to `developers`.

```
root@LAPTOP-KRQIMQLA:~# usermod -g developers lilly
root@LAPTOP-KRQIMQLA:~# id lilly
uid=1001(lilly) gid=1004(developers) groups=1004(developers),100(users),1005
(testers)
```

Level 3: File Ownership

18. Create a directory named `LinuxLab`.
19. Create a file named `test.txt` inside it.
20. Change the owner of `test.txt` to `pooja`.
21. Change the group owner of `test.txt` to `developers`.
22. Display the file's owner and group

```
root@LAPTOP-KRQIMQLA:~# mkdir -p /root/LinuxLab
root@LAPTOP-KRQIMQLA:~# touch /root/LinuxLab/test.txt
root@LAPTOP-KRQIMQLA:~# chown pooja /root/LinuxLab/test.txt
root@LAPTOP-KRQIMQLA:~# chgrp developers /root/LinuxLab/test.txt
root@LAPTOP-KRQIMQLA:~# ls -l /root/LinuxLab/test.txt
-rw-r--r-- 1 pooja developers 0 Oct 12 12:28 /root/LinuxLab/test.txt
```

Level 4: Symbolic Permissions (`chmod`)

23. Give the owner read, write, and execute permissions.
24. Add write permission to the group.
25. Add write permission to others.
26. Remove write permission from others.
27. Give the group read-only permission.
28. Set owner permissions to `rwx`, group permissions to none, and others permissions to none.
29. Set permissions to `u=rwx,g=rx,o=r`.

```
root@LAPTOP-KRQIMQLA:~# chmod u+rwx /root/LinuxLab/test.txt
root@LAPTOP-KRQIMQLA:~# chmod g+w /root/LinuxLab/test.txt
root@LAPTOP-KRQIMQLA:~# chmod o+w /root/LinuxLab/test.txt
root@LAPTOP-KRQIMQLA:~# chmod o-w /root/LinuxLab/test.txt
root@LAPTOP-KRQIMQLA:~# chmod g=r /root/LinuxLab/test.txt
root@LAPTOP-KRQIMQLA:~# chmod u=rwx,g=,o= /root/LinuxLab/test.txt
root@LAPTOP-KRQIMQLA:~# chmod u=rwx,g=rx,o=r /root/LinuxLab/test.txt
```

Level 5: Numeric Permissions (`chmod`)

30. Set file permissions to `777`.
31. Set file permissions to `755`.
32. Set file permissions to `644`.
33. Set file permissions to `700`.
34. Set file permissions to `750`.
35. Set file permissions to `640`.

```
root@LAPTOP-KRQIMQLA:~# chmod 777 /root/LinuxLab/test.txt
root@LAPTOP-KRQIMQLA:~# chmod 755 /root/LinuxLab/test.txt
root@LAPTOP-KRQIMQLA:~# chmod 644 /root/LinuxLab/test.txt
root@LAPTOP-KRQIMQLA:~# chmod 700 /root/LinuxLab/test.txt
root@LAPTOP-KRQIMQLA:~# ^C
root@LAPTOP-KRQIMQLA:~# chmod 750 /root/LinuxLab/test.txt
```

36. Convert `rwxr-xr-x` into numeric permissions.
37. Convert `rw-r--r--` into numeric permissions.
38. Convert `rwxr-x---` into numeric permissions.

`rwxr-xr-x` → `755`

`rw-r--r--` → `644`

`rwxr-x---` → `750`